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LAUREA IN INGEGNERIA DELL'AUTOMAZIONE

Hierarchical VLMs Planning and Control with Ambiguity resolution

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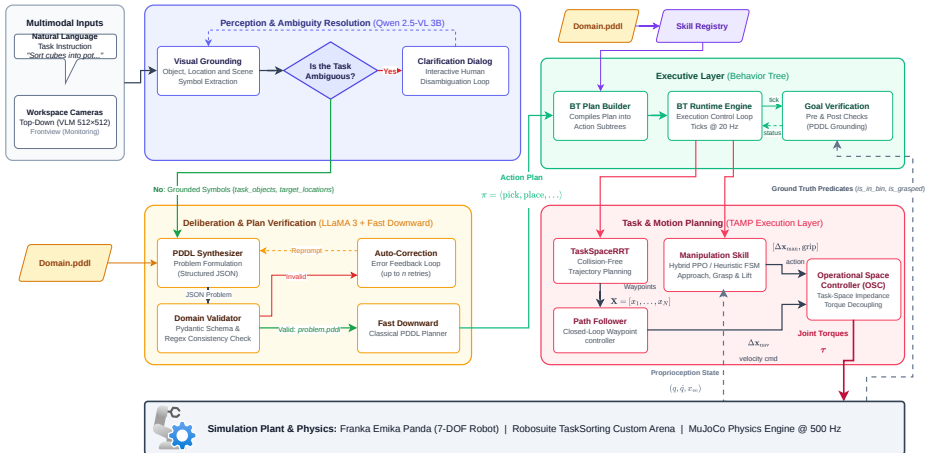
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Outline

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2. Executive Layer
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Neuro-Symbolic Planning architecture



Perception: Visual Grounding & Ambiguity Resolution

Grounds natural language human instructions in the observed scene and explicitly reasons about task ambiguity through multi-turn visual dialogue.

Disambiguation Pipeline

► Visual Scene Grounding (Qwen 2.5-VL):

Parses the camera frame into entity schemas:

- task_objects: Manipulable items (e.g., cans, cubes).
- target_locations: Receptacles (e.g., sorting bin, pot).

► Ambiguity Detection: Evaluates query against detected scene entities:

- Flags referential ambiguity (e.g., multiple cubes present).
- Schema: AmbiguityReasoning(task_ambiguous, question).

► Human-in-the-Loop Clarification: Prompts the user with a targeted question, appends response to chat history, and disambiguates the scene.

► Symbols Grounding: Outputs clean symbols {Objects, Locations} directly to the LLaMA-3 PDDL synthesizer.

Clarification Example

Ambiguous Human Prompt:

"Put the cube in the container."

Scene Detection (Qwen 2.5-VL):

- Objects: [yellow_cube, purple_cube]
 - Locations: [sorting_bin, pot]
- ⇒ **Ambiguity Flagged: True**

Robot: *"Which cube should I move, the yellow or the purple one? And into the pot or the bin?"*

Human: *"The yellow cube into the pot."*

Disambiguated Symbols for PDDL:

- Target: yellow_cube → (type obj)
- Destination: pot → (type location)

Key Benefit: Eliminates hallucinated assumptions before symbolic deliberation.

VLM Architectures: Dual-Encoder Foundations

Vision-Language Models (VLMs) integrate visual perception with linguistic reasoning by bridging continuous visual feature spaces and discrete language token spaces.

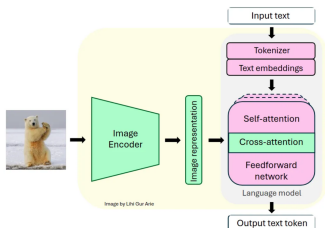


Figure: Standard Dual-Encoder VLM Architecture

Language Encoder (Transformer)

- ▶ Encoders transform sequences into numerical representation called **embeddings**.
- ▶ A **self-attention mechanism** allows focus on important tokens.
- ▶ **Decoders** output the most statistically probable output.

Vision Encoder (CNN / ViT)

- ▶ Typically utilizes Convolutional Neural Networks (CNNs) or Visual Transformers (ViTs).
- ▶ A ViT processes an image into patches and treats them as sequences, akin to tokens.

$$\text{Image } \mathbf{I} + \text{Text } \mathbf{T} \xrightarrow{\text{VLM}} \text{Structured Output } \mathbf{Y}$$

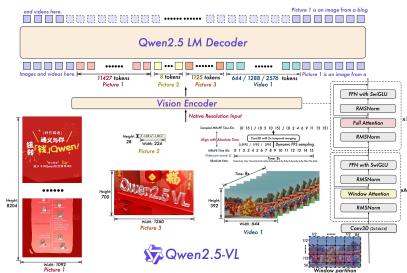
VLM Training methodologies

It is the mechanism that allow fusing information from both vision and language encoders so to correlate images with text and make decision on the 2 modalities together:

- ▶ Contrastive learning: uses joint or shared embedding space. The VLM is trained on datasets of image-text pairs and learns to minimize the distance between embeddings of matching pairs and maximize it for nonmatching pairs.
- ▶ Masking: models learn to predict randomly obscured parts of an input text or image.
- ▶ Generative model training: entails learning to generate new data
- ▶ Pretrained models: A pretrained LLM and a pretrained vision encoder can be used, with an added mapping network layer that aligns or projects the visual representation of an image to the LLM's input space.

Qwen 2.5-VL Architecture

- ▶ Qwen2.5 LM Decoder
- ▶ ViT:3D convolution to split input visual data into a series of 14*14 patches
- ▶ process video info: preprocessor for dynamic image size adjustment, dynamic frame extraction



Why this choice?

- ▶ **Dynamic Resolution (Naive ViT):** Preserves small object aspect ratios without distortion.
- ▶ **Local Real-Time Inference:** Runs locally via Ollama (1.25 s latency, zero API dependency).
- ▶ **High Spatial Fidelity:** Superior grounding of small manipulable items and receptacle boundaries.

Symbolic Deliberation: PDDL Formulation

Decouples domain physics from task specification using **PDDL** (Planning Domain Definition Language), guaranteeing causal correctness before robot motion.

Formal Planning Tuple

$$\mathcal{P} = (\mathcal{D}, s_0, G)$$

► Planning Domain $\mathcal{D} = (\mathcal{L}, \mathcal{A})$ [Static]:

- $\mathcal{L}$: Predicates describing state relations.
- $\mathcal{A}$: Action schemas with preconditions and effects.
- *Fixed and verified offline* (robot capability bounds).

► Planning Problem $\mathcal{P}$ [Dynamic]:

- s_0 : Completely grounded initial state from VLM.
- G : Target goal conditions from user task.
- *Synthesized online* via LLaMA 3 from perception.

Key Principle: Domain physics are fixed; only state and goal are queried online.

Manipulation Domain

Grounded State Predicates $\mathcal{L}$:

- `(on ?obj ?loc)`: Object placed on table/bin/pot.
- `(holding ?obj)`: Robot gripper contact established.
- `(on-table ?obj)`: Object resting on tabletop.

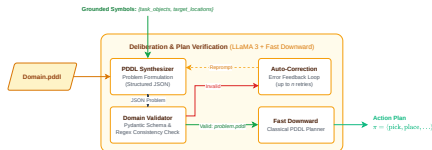
Robotic Action Schemas $\mathcal{A}$:

- `pick(?o)`:
 - Pre: `(on-table ?o), (not (holding ?o))`
 - Eff: `(holding ?o), (not (on-table ?o))`
- `place(?o, ?loc)`:
 - Pre: `(holding ?o)`
 - Eff: `(on ?o ?loc), (not (holding ?o))`

PDDL (Planning Domain Definition Language) is the standard for Classical AI Planning. To solve planning problems, two main paradigms exist: **State Space Search** and **Plan Space Search**.

Plan Verification: Fast Downward Solver

Fast Downward acts as the symbolic deliberation engine, translating VLM-grounded initial states into verified action sequences via **Heuristic Forward State-Space Search** in < 0.08 s.



Forward State-Space Search (Progression)

- **Search Paradigm:** Operates in state space S (complete valuations), progressing forward: $s_0 \xrightarrow{a_1} s_1 \rightarrow \dots \rightarrow s_g \models G$.
- **Why Forward Search?** VLM provides a *completely grounded initial state* s_0 , making progression directly evaluable.
- **State Transitions:** Applies applicable action a ($pre(a) \subseteq s$):
 $s' = (s \setminus del(a)) \cup add(a)$.
- **Neuro-Symbolic Verifier:** Deterministically validates causality, eliminating LLM hallucinations.

Heuristic Search Engine (lama-first)

- **Lazy GBFS:** Greedy best-first search deferring node evaluation for high-speed plan discovery.
- **Fast-Forward (h_{FF}):** Delete-relaxation on Relaxed Planning Graph ($del(a) = \emptyset$), estimating remaining action steps.
- **Landmark Count (h_{LM}):** Tracks causal landmarks (facts/actions mandatory in every valid plan).
- **Preferred Operators:** Prunes search branching by prioritizing helpful actions from h_{FF} and h_{LM} .

PDDL Problem Generation Experiments

Autonomous synthesis of PDDL Problem files $\mathcal{P} = (\mathcal{D}, s_0, G)$ from natural language instructions and VLM scene grounding, benchmarked across 28 tabletop simulation trials.

Generation, Failures & Auto-Repair

- ▶ **Structured Formulation:** LLaMA 3 outputs typed JSON validated by Pydantic (Problem, PDDLObject, Predicate) from Qwen 2.5-VL scene groundings.
- ▶ **Observed Failure Modes (Raw LLM):**
 - ▶ *Predicate Hallucination:* Generates (in ?x ?bin) instead of domain-defined (on ?x ?loc)
 - ▶ *Type Mismatches:* Target fixtures classified as obj instead of location.
- ▶ **Automated Feedback Loop:** Programmatic validator parses errors and reprompts LLaMA 3 with exact failure traces (max 5 retries).
- ▶ **Auto-Healing:** Inferred type assignment for uninstantiated receptacle locations.

Empirical Benchmark (28 Trials)

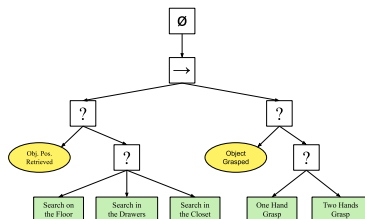
Evaluation Metric	Result
Logged Benchmark Trials	28
1st-Pass Solver Feasibility	26 / 28 (92.9%)
Success after Feedback Loop	28 / 28 (100%)
Fast Downward Parser Errors	0%
Avg. Symbolic Solve Time	0.084 s
Max Sequential Objects	4 (8 plan steps)

Key Takeaway for TAMP

- ▶ **Formal Gatekeeper:** Fast Downward catches unreachable goals *before* physical execution.
- ▶ **Execution Reliability:** Completely shields the Behavior Tree from malformed robotic primitives.

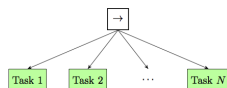
Executive Layer: Behavior Trees

- ▶ A **Behavior Tree** is a directed tree with nodes classified as root, control flow, or execution (tasks).
- ▶ Execution starts from the root, which sends *ticks* to its children at a set frequency.
- ▶ A node returns one of three statuses: **Running**, **Success**, or **Failure**.



Control Flow Nodes

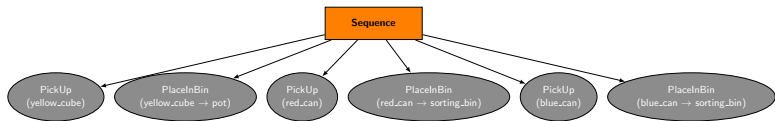
- ▶ **Selector:** A fallback node that executes the first child that does not fail. Returns **Success** or **Running**.
- ▶ **Sequence:** Executes children in order until one succeeds. Returns **Failure** or **Running** immediately if a child fails or is still running.



How to build Behavior Trees: `py_trees` library

`py_trees` is a Python library used to construct Behavior Trees efficiently. It relies primarily on two base classes: **Behaviours** and **Composites**.

- ▶ **Behaviour (Leaf):** The smallest element in a tree. Behaviours typically represent a **condition** (e.g., *Am I hungry?*) or an **action** (e.g., *Buy chocolate cookies*).
- ▶ **Composites (Control Flow):** Responsible for directing the path traced through the tree during a tick.
 - ▶ **Selectors:** Fallback decision-makers.
 - ▶ **Sequences:** Order-dependent factories.
 - ▶ **Parallels:** Concurrent execution factories.



BT Builder: Parsing and Sequence Compilation

1. Parsing PDDL Plan

(`parse_pddl_action`) Cleans raw planner text into structured Python data:

- ▶ **Filters Clutter:** Discards comments starting with ; (e.g., ; cost = 2) and blank lines.
- ▶ **Strips Planner Syntax:** Removes step prefixes (e.g., 0:) and outer parentheses (...).
- ▶ **Extracts Tokens:** Normalizes action name to lowercase and collects arguments:

```
"0: (place yellow_cube pot)" → ("place",  
    ["yellow_cube", "pot"])
```

2. Plan to Sequence

(`pddl_plan_to_sequence`) Converts the parsed action list into an executable Behavior Tree Sequence:

- ▶ **Skill Dispatch:** Uses `SKILL_REGISTRY` to map actions to robot skills ("pick", "place").
- ▶ **Argument Binding:** Passes simulator environment and object targets (object, asset).

Compilation Pipeline Example

1. Planner Output (Fast Downward):

```
; plan generated by fast downward  
0: (pick yellow_cube)  
1: (place yellow_cube pot)  
; cost = 2
```

⇓ *parsed & compiled into BT*

2. Executable PyTrees Sequence:

```
[~] pddl_plan_sequence (memory=True)  
|-- PickUp(object='yellow_cube')  
'-- PlaceInBin(object='yellow_cube',  
    asset='pot')
```

Design Rationale

Cleanly decouples **symbolic task planning** (Fast Downward) from the **physical robot control** (Motion planning).

BT Builder: Goal Checking and Reactive Tree Generation

3. Goal-Guarded Wrapper

(wrap_with_goal_check) A plain sequence is open-loop. If an object drops, it fails blindly. We wrap it in a reactive fallback loop:

- ▶ **Pre-Check (GoalCheck_Pre)**: Verifies if goal is already met before moving. If yes → succeeds immediately (no redundant actions).
- ▶ **Selector (memory=False)**: Re-evaluates continuously on every tick for dynamic preemption.
- ▶ **Post-Check (GoalCheck_Post)**: Confirms physical completion after the sequence finishes.
- ▶ **Retry Loop (Retry)**: If the grasp slips and post-check fails, automatically retries the plan up to N times.

4. Main Entrypoint

(build_bt_from_pddl_plan) Single builder API providing two execution modes:

- ▶ **wrap_goal_check=False**: Raw Sequence (open-loop baseline).
- ▶ **wrap_goal_check=True**: Self-healing tree with retry loop.

Goal-Guarded Tree Architecture

```
[?] goal_guarded_root (Selector)
|-- [>] GoalCheck_Pre (already met?)
'-- [R] retry_until_goal (max=10)
    '-- [->] attempt (Sequence)
        |-- [->] pddl_plan_sequence
        |   |-- PickUp(yellow_cube)
        |   '-- PlaceInBin(yellow_cube->pot)
        '-- [>] GoalCheck_Post (verified?)
```

Failure Recovery Example (Cube Drop)

1. Robot picks up cube, but gripper slips mid-air.
2. Sequence finishes → GoalCheck_Post checks pot.
3. Cube is missing → returns FAILURE.
4. Retry catches failure → re-runs the plan → task completed successfully!

BT Nodes: Skills & Conditions

Action Skills (`skills.py`)

Subclasses of `py_trees.behaviour.Behaviour`:

- ▶ **Non-Blocking Stepping:** Returns `RUNNING` while the robot moves, and `SUCCESS` or `FAILURE` upon completion.
- ▶ **TaskSpaceRRT Integration:** Computes collision-free $SE(3)$ end-effector trajectories dynamically for the 7-DOF Panda arm.
- ▶ **Operational Primitives:**
 - ▶ `MotionPlanningPickUp`: Approach → Grasp → Lift.
 - ▶ `MotionPlanningPlaceInBin`: Transit → Lower → Release → Retreat.

Condition Nodes (`conditions.py`)

Dual verification paradigm:

- ▶ **PDDLGoalCheck (Symbolic Grounding):**
 - ▶ Parses goal formulas from `problem.pddl`.
 - ▶ Evaluates simulator predicates deterministically (`is_in_bin`, `is_grasped`).
 - ▶ Fast, deterministic, and zero-latency.
- ▶ **VisualCheck (VLM Inspection):**
 - ▶ Adopts Wake et al. self-prompting.
 - ▶ Queries VLM (`vlm.query_fn`) on camera frames for open-world semantic validation.

Reactive Execution & Runtime Tree

Runtime Execution Flow

1. **Initial Check:** GoalCheck_Pre runs. If goal holds, the tree returns SUCCESS immediately (no redundant actions).
2. **Execution:** On failure, the Fallback ticks `retry_until_goal` → `pddl_plan_sequence`.
3. **Closed-Loop Motion:** Each skill executes collision-free RRT trajectories at 20 Hz.
4. **Verification & Recovery:** GoalCheck_Post validates final state. If an object is dropped, the Retry decorator triggers re-execution.

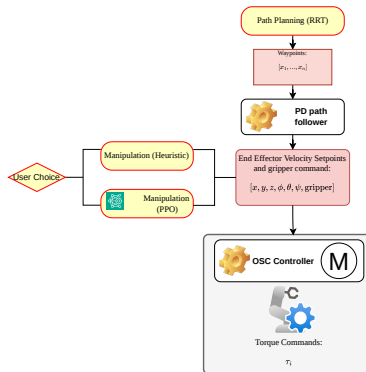
Generated Behavior Tree Structure

```
[?] goal_guarded_root (Selector)
|-- [>] GoalCheck_Pre
'-- [R] retry_until_goal (max=3)
    '-- [->] attempt (Sequence)
        |-- [->] pddl_plan_sequence
        |   |-- PickUp(yellow_cube)
        |   |-- PlaceInBin(yellow_cube->pot)
        |   |-- PickUp(red_can)
        |   '-- PlaceInBin(red_can->bin)
        '-- [>] GoalCheck_Post
```

Reactive Layer: Motion and Path Planning

Reactive Layer Roles

- ▶ **Symbolic-to-Continuous Grounding:** Translates discrete BT skills (PickUp, PlaceInBin) into continuous robot motion.
- ▶ **Perception:** Accounts for physical geometry, obstacles, and contact dynamics ignored by PDDL.
- ▶ **Motor Control:** Operates at 20 Hz with real-time state feedback from MuJoCo Simulator Environment.



Hierarchical Motion Decomposition

- ▶ **1. Path Planning (Task Space RRT):** Solves pure geometric obstacle avoidance in free space ($\mathbb{R}^3$).
- ▶ **2. Trajectory Tracking:** Closed-loop proportional tracking of Cartesian waypoints $[x_1, \dots, x_n]$ generated by the RRT.
- ▶ **3. Manipulation (PPO / Heuristic):** Governs fine contact dynamics, compliant grasping, and lifting.
- ▶ **4. Low Level Controller (OSC):** Takes as input the cartesian velocities from the Tracking and Manipulation modules and output the joint torques τ_i needed to control the robot.

Path Planning: RRT

Rapidly Random Trees (RRT) is a Probabilistic Path Planning method used to efficiently construct a path in continuous spaces.

Algorithm Goal Oriented RRT Algorithm

```
1:  $G.\text{init}(x_{\text{start}})$ 
2: for  $i = 1 \dots K$  do
3:    $p = \text{UniformRandom}(0, 1)$ 
4:   if  $p \leq p_{\text{goal}}$  then
5:      $x_{\text{rand}} \leftarrow x_{\text{goal}}$ 
6:   else
7:      $x_{\text{rand}} \leftarrow \text{RandPose}()$ 
8:   end if
9:    $x_{\text{nearest}} \leftarrow \text{NearestVertex}(x_{\text{rand}}, G)$ 
10:   $x_{\text{new}} \leftarrow \text{NewConf}(x_{\text{nearest}}, x_{\text{rand}}, \Delta x)$ 
11:  if  $\text{CollisionFree}(x_{\text{nearest}}, x_{\text{new}})$  then
12:     $G.\text{add\_vertex}(x_{\text{new}})$ 
13:     $G.\text{add\_edge}(x_{\text{nearest}}, x_{\text{new}})$ 
14:    if  $x_{\text{new}} \in X_{\text{goal}}$  then return Path Found
15:  end if
16: end if
17: end for
18: return  $G$ 
```

The goal oriented version samples the goal point with a given probability p while a random point with probability $1 - p$.

Trajectory Tracking

- ▶ **Cartesian Setpoints:** The waypoints generated by the task-space RRT are Cartesian poses used as setpoints for a **Proportional (P) Controller**.
- ▶ **Tracking Error:** At each time step, the position error is defined as:

$$\mathbf{e} = \mathbf{p}_{\text{target}} - \mathbf{p}_{\text{EF}}$$

- ▶ **Control Law:** The controller outputs velocity commands bounded by a maximum speed:

$$\mathbf{v}_{\text{cmd}} = \text{clip}(K_p \mathbf{e}, -v_{\text{max}}, v_{\text{max}})$$

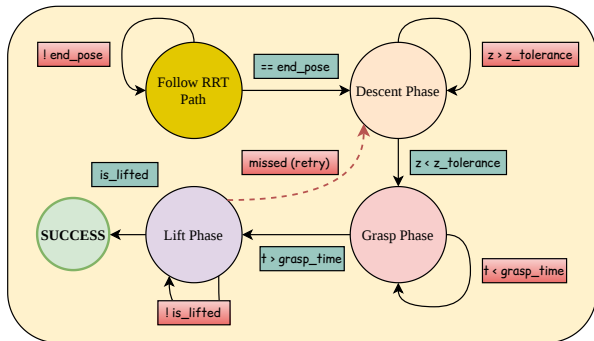
which are sent to the underlying operational space controller.

- ▶ **Application:** Used specifically during the **approach phase** in the pick action and during the **place action**. The place skill itself is completely based on the RRT planner and the trajectory tracker, while the pick skill is composed by more modules and need further explanation.

Manipulation: Heuristic Approach

Consists of a **Staged FSM**, that executes the following phases:

1. **RRT Planning:** Traverse collision-free waypoints to hover pose ($== \text{end_pose}$).
2. **Descent Phase:** Descend vertically until reaching grasp height ($z \leq z_{\text{tol}}$).
3. **Grasp Phase:** Actuate gripper firmly for set duration ($t > t_{\text{grasp}}$).
4. **Lift Phase:** Elevate object to safe height to verify pick. If it fails, it will retry from the Descent phase.



Proximal Policy Optimization

Proximal Policy Optimization is an on-policy actor-critic Reinforcement Learning method. PPO updates policy parameters θ_{k+1} by maximizing the **Clipped Surrogate Objective**:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where the probability ratio is defined as:

$$r_t(\theta) = \frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_k}(a_t | s_t)}$$

- ▶ **Clipped Objective:** The hyperparameter $\epsilon \approx 0.2$ penalizes moving $r_t(\theta)$ outside $[1 - \epsilon, 1 + \epsilon]$, preventing destructively large policy updates.
- ▶ **Actor-Critic Architecture:** Jointly optimizes a policy network $\pi_{\theta}(a|s)$ and a state-value baseline $V_{\phi}(s)$ using Generalized Advantage Estimation (GAE).

Manipulation: PPO Approach

TaskSpaceRRT provides coarse global navigation to a pre-grasp hover pose (~ 18 cm above the object). The **PPO policy** then takes over using a minimal, translation-invariant **11D observation vector**:

$$\mathbf{o}_t = \left[\underbrace{\Delta \mathbf{p}}_{\mathbb{R}^3}, \underbrace{\mathbf{q}_{\text{grip}}}_{\mathbb{R}^2}, \underbrace{\dot{\mathbf{q}}_{\text{grip}}}_{\mathbb{R}^2}, \underbrace{z_{\text{rel}}^{\text{EEF}}}_{\mathbb{R}^1}, \underbrace{\Delta z_{\text{obj}}}_{\mathbb{R}^1}, \underbrace{c_{\text{grasp}}}_{\{0,1\}}, \underbrace{m_{\text{geom}}}_{\{0,1\}} \right]^T \in \mathbb{R}^{11}$$

Kinematics & Spatial Relations

- ▶ $\Delta \mathbf{p} = \mathbf{p}_{\text{obj}} - \mathbf{p}_{\text{EEF}}$: 3D relative vector (translation invariance across table).
- ▶ $z_{\text{rel}}^{\text{EEF}} = p_{\text{EEF},z} - z_{\text{table}}$: Table clearance to avoid surface collisions.
- ▶ $\Delta z_{\text{obj}} = p_{\text{obj},z} - p_{\text{obj},z}^{(0)}$: Vertical displacement measuring lift progress.

Gripper State & Morphology

- ▶ $\mathbf{q}_{\text{grip}}, \dot{\mathbf{q}}_{\text{grip}}$: Left/right finger joint positions and velocities (contact stall detection).
- ▶ $c_{\text{grasp}} \in \{0, 1\}$: Bilateral force-closure flag (Robosuite contact normal check).
- ▶ $m_{\text{geom}} \in \{0, 1\}$: Shape token (0: cube, 1: can) enabling multi-geometry conditioning.

Manipulation: Shaped Reward Formulation

The agent is trained with a multi-stage continuous-discrete reward guiding the transition from approach $\rightarrow$ grasp $\rightarrow$ lift:

$$R_t = r_{\text{reach}} + r_{\text{grip}} + r_{\text{grasp}} + r_{\text{lift}} - c_{\text{step}} + r_{\text{terminal}}$$

Continuous Shaping Terms

- ▶ **Smooth Reaching** ($d = \|\Delta \mathbf{p}\|_2$):
 $r_{\text{reach}} = 1.0 - \tanh(10.0 \cdot d)$
- ▶ **Grasp Maintenance:**
 $r_{\text{grasp}} = 2.0 \cdot \mathbb{I}(\text{grasped})$
- ▶ **Conditioned Vertical Lift:**
 $r_{\text{lift}} = 20.0 \cdot \max(0, \Delta z_{\text{obj}}) \cdot \mathbb{I}(\text{grasped})$
(Avoids lifting with an empty gripper)
- ▶ **Alive Cost:** $c_{\text{step}} = 0.01$ (incentivizes speed)

Action Shaping & Terminals

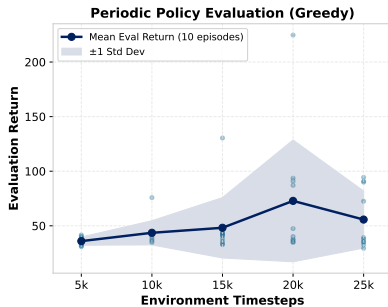
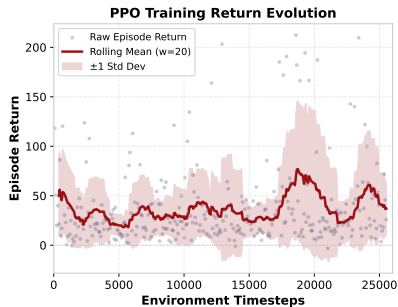
- ▶ **Gripper Timing Incentive** r_{grip} :

$$r_{\text{grip}} = \begin{cases} +0.5 & \text{if } d < 4 \text{ cm} \wedge a_g > 0 \\ -0.25 & \text{if } d < 4 \text{ cm} \wedge a_g \leq 0 \\ -0.2 & \text{if } d \geq 4 \text{ cm} \wedge a_g > 0.2 \\ 0 & \text{otherwise} \end{cases}$$

Prevents premature closure before alignment.

- ▶ **Terminal Sparse Rewards:**
 $+15.0$ if $\Delta z_{\text{obj}} \geq 6 \text{ cm} \wedge \text{grasped}$
(Success)
 -2.0 if $p_{\text{obj},z} < z_{\text{table}} - 4 \text{ cm}$ (Drop)

Manipulation: PPO Training & Evaluation



Training Progression (25k Steps)

- **Exploration Phase:** High variance in initial rollouts as the policy explores grasp stability.
- **High-Return Peaks (> 150):** Successful grasp + vertical lift + sparse completion bonus (+15).

Deterministic Evaluation

- Evaluated every 5,000 steps over 10 test rollouts.
- Greedy policy achieves consistent ascent, climbing from initial hover (~ 36) to peak performance (~ 73).

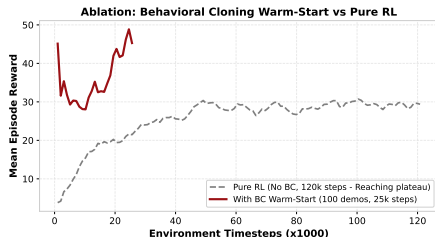
Manipulation: Why Behavioral Cloning Warm-Start?

Algorithmic Oracle Demonstrations

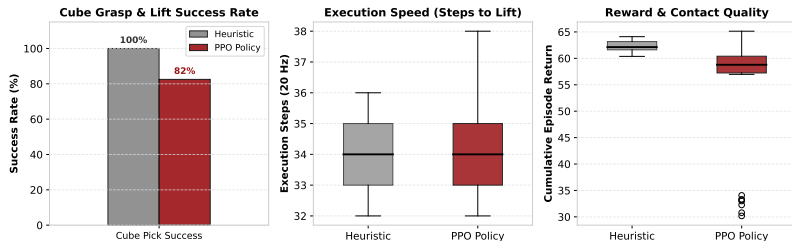
- ▶ Generated via a closed-loop 3-stage state-machine using simulation ground-truth poses.
- ▶ 100 demos ($\sim 3,000$ transitions) pre-trained for 150 epochs via supervised MSE loss.

Key Empirical Findings

- ▶ **Pure RL (120k steps):** Gets stuck in the "reaching local minimum" (≈ 29 reward), never discovering the precise descent-grasp-lift sequence.
- ▶ **BC Warm-Start (25k steps):** Initializes policy in the basin of attraction, achieving immediate task competence in $5\times$ fewer steps.



Manipulation: Heuristic vs PPO Comparison



Cube Manipulation Performance

- ▶ **High Reliability:** PPO achieves **82.5%** success rate (vs. 100% analytical oracle) using local relative observations.
- ▶ **Micro-Trajectory Parity:** Identical physical motion speed: **1.71 s** (34.1 steps) vs **1.70 s** (34.0 steps).
- ▶ **Eliminates Dwell Timers:** The heuristic forces a conservative **2.0 s** (40 ticks) pause before lifting; PPO lifts instantly upon contact.

Trade-Offs & Hybrid Synergy

- ▶ **Dynamic Compliance:** PPO modulates continuous velocities based on real-time force closure rather than rigid fixed gains ($K_p = 10$).
- ▶ **Privileged vs Reactive:** Heuristic requires exact object coordinates; PPO relies only on local $\Delta \mathbf{p}$ and joint kinematics.
- ▶ **Safety Guard:** Hybrid pick node falls back to heuristic if step budget (≤ 90) expires.

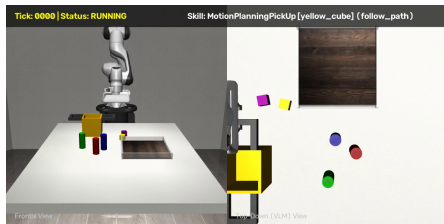
Experimental Setup: Simulation Environment

Robosuite TaskSorting Arena

- ▶ **Robot:** 7-DoF Franka Emika Panda with parallel-jaw gripper (20 Hz OSC control).
- ▶ **Physics:** MuJoCo headless physics engine with compliant contact dynamics.

Sensory & Tabletop Assets

- ▶ **Dual Viewpoints:** Orthographic *top-down* (VLM grounding) + perspective *frontview* (monitoring).
- ▶ **Objects:** 5 pickable items (yellow/purple cubes, red/blue/green cans).
- ▶ **Containers:** Sorting bin (30×30 cm) + handled pot (collision obstacles).



End-to-End TAMP Pipeline Execution

Task Mission: *"Put the red can in the sorting bin, then put the yellow cube in the pot, then put the blue can in the sorting bin."*

1. Perception & Plan

- ▶ **Qwen 2.5-VL:** Resolves object locations from top-down frame.
- ▶ **Llama 3:** Synthesizes initial/goal PDDL predicates.
- ▶ **Fast Downward:** Generates 6-step symbolic action plan in < 0.1 s.

2. BT Compilation

- ▶ Compiled into reactive `py_trees` structure.
- ▶ Encapsulates pre-conditions, action sequences, and fallback recovery nodes.
- ▶ Continuous state tick monitoring at 20 Hz.

3. Motion & Control

- ▶ **TaskSpaceRRT:** Computes collision-free transit paths (~ 30 ms).
- ▶ **Hybrid PPO:** Compliant grasp and lift (≈ 1.7 s).
- ▶ **Placement:** Waypoint tracking to bin/pot.

Experimental Results: Subsystem Performance

Quantitative evaluation across end-to-end multi-object sorting missions:

Pipeline Layer	Module / Algorithm	Mean Latency	Success Rate
Visual Grounding	Qwen 2.5-VL (3B)	1.25 s	100%
PDDL Formulation	Llama 3 (8B)	1.80 s	100%
Symbolic Planning	Fast Downward (A^*)	0.07 s	100%
Motion Planning	TaskSpaceRRT	0.03 s	100%
Manipulation (Pick)	Hybrid PPO Policy	1.71 s	88% (PPO)
Fallback Execution	Deterministic Staged	1.70 s	100% (Fallback)
End-to-End System	Complete Architecture	Total Mission	100%

Key Findings

- ▶ Symbolic & motion planning combined require < 0.1 s, enabling agile replanning.
- ▶ Zero collisions observed with container walls across all transit trajectories.

Execution Guarantees

- ▶ The hybrid RL-fallback mechanism ensures zero mission halts on manipulation failures.
- ▶ Modular design decouples high-level task planning from local contact mechanics.

Conclusions & Future Perspectives

A hierarchical 3-layer architecture bridging multimodal foundation models with formal symbolic deliberation and closed-loop reactive manipulation.

Core Contributions

- ▶ **Neuro-Symbolic Deliberation:** Autonomous extraction of (s_0, G) from RGB frames and language, fortified by interactive disambiguation and auto-repair (100% solver feasibility).
- ▶ **Formal Fast Downward Gatekeeper:** Guarantees causal plan validity in < 0.08 s before physical dispatch.
- ▶ **Reactive Executive Layer:** Behavior Trees compile action sequences into tick-based subtrees with continuous pre/post-condition monitoring.
- ▶ **Robust Hybrid Control:** TaskSpaceRRT ensures collision-free transit, while BC-bootstrapped PPO achieves reliable contact-rich grasping.

Future Developments

- ▶ **Domain & Problem Co-Synthesis:** Moving beyond fixed domain templates toward online predicate invention and automated action schema discovery.
- ▶ **Skill Registry Augmentation:** Expanding the primitive library to articulated objects and different actions.
- ▶ **Unified RL/IL for TAMP:** End-to-end hierarchical policy learning bridging task logic with real-time visuomotor control.
- ▶ **Sim-to-Real Transfer:** Hardware validation on physical 7-DOF Franka Emika Panda setups.

Key References



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